PRODUCT USER MANUAL

Sea Surface Temperature Ocean Monitoring Indicator for the European North West Shelf Seas :

OMI_CLIMATE_SST_NORTHWESTSHELF_area_averaged_anomalies
OMI_CLIMATE_SST_NORTHWESTSHELF_trend

Issue 3.0

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GLOSSARY AND ABBREVIATIONS

MOI	Mercator Ocean International
NetCDF	Network Common Data Form
OMI	Ocean Monitoring Indicator
PUM	Product User Manual
QUID	Quality Information Document
SST	Sea Surface Temperature
REP	Reprocessed

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DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

I INTRODUCTION

I.1. Summary

This document is the user manual for the Copernicus Marine Sea Surface Temperature (SST) Ocean Monitoring Indicator (OMI) for the European North-West Shelf Seas (NWS). The NORTHWESTSHELF SST OMI belongs to the ocean temperature and salinity OMI family and provides a 1D time series of area averaged monthly mean SST anomalies and a 2D map of SST trend for the NWS Seas. SST anomalies are computed with respect to the 1991-2020 reference period (climatology). The NWS SST OMI is built from the CMEMS Reprocessed European North West Shelf Iberia-Biscay-Irish areas(REP ATL) SST L4 product[1] (see PUM[2] and QUID[3]).
Specifically, the NORTHWESTSHELF SST OMI includes two products/datasets (here, product name coincides with dataset name):

- OMI_CLIMATE_SST_NORTHWESTSHELF_area_averaged_anomalies, which provides monthly mean anomaly time series of SST averaged over the area;
- OMI_CLIMATE_SST_NORTHWESTSHELF_trend, which contains the 2D map of annual trend over the area.

The NORTHWESTSHELF SST OMI products are published on the Copernicus Marine dissemination server after automatic and delayed-mode quality controls. These OMI products are available online and disseminated through the CMEMS Information System. Files downloaded are in NetCDF format and follow CF-1.7 convention.

1 SST_ATL_SST_L4_REP_OBSERVATIONS_010_026

2 CMEMS Ocean Monitoring Indicators Guideline (copernicus.eu)

3 CMEMS-SST-QUID-010-026.pdf (copernicus.eu)

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.

The NORTHWESTSHELF SST OMI products are described in the Quality Information Document (QUID) https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-OMI-QUID-NORTHWESTSHELF-SST.pdf.

I.2. History of changes

February 2019	Publication of NORTHWESTSHELF SST OMI (OMI_CLIMATE_SST_NORTHWESTSHELF_area_averaged_anomalies; OMI_CLIMATE_SST_NORTHWESTSHELF_trend)
September 2020	Update of the upstream reference dataset and addition of the year 2019
September 2021	Update of the upstream reference dataset and addition of the year 2020
September 2022	Update of the upstream reference dataset and addition of the year 2021
November 2022	Update of the upstream reference dataset and addition of the year 2022
November 2024	Update of the upstream reference dataset, addition of the year 2023, and a new climatology from 1991 to 2020

II DESCRIPTION OF THE PRODUCT

II.1. General information

II.1.1. OMI_CLIMATE_SST_NORTHWESTSHELF_area_averaged_anomalies

Product(s) name	OMI_CLIMATE_SST_NORTHWESTSHELF_area_averaged_anomalies		
Geographical coverage	European North West Shelf Seas		
Variables	Time series of SST anomalies		
Update frequency	Annual[4]		
Available time series	1982-01-01 to 2023-12-31		
Delivery mechanism	Marine Data Store		
Temporal resolution	Monthly averages		
Integration depth	Surface		
Format	NetCDF CF1.7		

I.1.1. OMI_CLIMATE_SST_NORTHWESTSHELF_trend

Product(s) name	OMI_CLIMATE_SST_NORTHWESTSHELF_trend		
Geographical coverage	European North West Shelf Seas		
Variables	Map of SST trend		
Update frequency	Annual[4]		
Available time series	1982-01-01 to 2023-12-31		
Delivery mechanism	Marine Data Store		
Temporal resolution	Time integrated		
Integration depth	Surface		
Format	NetCDF CF1.7		

I.2. Details of the datasets

4 In the present version , the update frequency is annual. However, the NORTHWESTSHELF SST OMI products update will change during the next releases of CMEMS, following the REP ATL L4 product update frequency.

For each product, one table with details of datasets content. For more details about each dataset, see the QUID (https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-OMI-QUID-NORTHWESTSHELF-SST.pdf)

I.2.1. OMI_CLIMATE_SST_NORTHWESTSHELF_area_averaged_anomalies

OMI_CLIMATE_SST_NORTHWESTSHELF_area_averaged_anomalies	
cmems_omi_climate_sst_northwestshelf_area_averaged_anomalies	Contains the 1D monthly mean SST anomaly time series averaged over the NORTHWESTSHELF and derived from the CMEMS REP European North West Shelf / Iberai Biscay Irish Seas L4 product (SST_NORTHWESTSHELF_SST_L4_REP_OBSERVATIONS_010_026). Anomalies are computed with respect to the 1991-2020 reference period. The filtered anomaly is obtained by using the X11-seasonal adjustment procedure.
	sst_anomaly [degree C] area averaged sea surface temperature anomaly sea_surface_temperature

I.2.2. OMI_CLIMATE_SST_NORTHWESTSHELF_trend

OMI_CLIMATE_SST_NORTHWESTSHELF_trend	
cmems_omi_climate_sst_northwestshlef_trend	Contains the 2D map of annual SST trend over the NORTHWESTSHELF and derived from the CMEMS REP European North West Shelf / Iberai Biscay Irish Seas L4 product (SST_ATL_SST_L4_REP_OBSERVATIONS_010_026).
	sst_trends [°C] surface temperature trend change_over_time_in_sea_surface_temperature

I.3. Product System Description

I.3.1. OMI_CLIMATE_SST_NORTHWESTSHELF_area_averaged_anomalies

This OMI includes SST anomalies, given as monthly mean time series starting on 1982 and averaged over the NORTHWESTSHELF. This OMI is derived from the Copernicus Marine REP ATL L4 SST product (SST_ATL_SST_L4_REP_OBSERVATIONS_010_026), which provided the SSTs used to compute the evolution of SST anomalies over the European North West Shef seas. Anomalies are computed against the 1991-2020 reference period.

I.3.2. OMI_CLIMATE_SST_NORTHWESTSHELF_sst_trend

This OMI includes the trend for the NORTHWESTSHELF, i.e. the rate of change per year. Trend computation is based on the X-11 seasonal adjustment procedure. This OMI is derived from the CMEMS REP ATL L4 SST product (SST_ATL_SST_L4_REP_OBSERVATIONS_010_026).

II FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

III FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this articles " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

III.1. Nomenclature of files

The NORTHWESTSHELF SST OMI files nomenclature when downloaded through the CMEMS Marine Data Store Services is based on the following nomenclature:

<datasetname>_<timestep>_P<productiontime>.nc

Where :

- **<datasetname>** is the dataset reference
- **<timestep>** is the first time step contained in the file**.**
- **<productiontime>** is the production time when the file was generated.
- **.nc** is the standard NetCDF filename extension.

Example:
cmems_omi_climate_sst_northwestshelf_area_averaged_anomalies_19820101_20231231_R19912020_P20240311.nc

IV REFERENCES

See the Quality Information Document (QUID) https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-OMI-QUID-NORTHWESTSHELF-SST.pdf